

THARUN KUMAR KSHEERASAGAR

Senior Architect

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SUMMARY

System Architecture and Hardware Simulation Engineer with over a decade of experience driving pre-silicon validation and accelerating compute workloads at NVIDIA, AMD, and Samsung. Expertise in hardware-software co-design, building optimized hybrid simulation frameworks, and resolving complex system-level bottlenecks to drastically reduce design turnaround times. Recent work focuses on the advanced AI infrastructure ecosystem, from spearheading the integration of NVIDIA iGPUs into MediaTek SoCs to driving architectural modeling for next-generation AI inference accelerators.

WORK EXPERIENCE

Stealth AI Hardware Startup

Hyderabad, India

Senior Architect

Apr 2026 – Sept 2026

- Developed a configurable SystemC performance model delivering exact cycle-by-cycle execution for a Groq-inspired AI inference accelerator card, establishing predictable, low-latency baseline metrics for real-time Transformer models while serving as a golden reference model for the compiler team
- Engineered an intuitive data-flow visualization tool to track real-time packet movement across selected hardware blocks, significantly accelerating cross-functional debug turnaround times for the compiler team.

NVIDIA

Hyderabad, India

Senior Architect

Jan 2025 – Apr 2026

- Led system-level validation and cross-functional debug for NVIDIA's next-generation N2X AI platform.
- Spearheaded the complex integration of an NVIDIA iGPU into a MediaTek SoC, achieving initial pre-silicon bring-up on a Virtual Platform in an accelerated 4-month timeframe.

AMD

Hyderabad, India

Member of Technical Staff Software Engineer

June 2019 – Jan 2025

- Designed and delivered a custom multi-threaded simulation kernel within a strict six-month timeframe, maintaining SystemC semantics while accelerating simulation speeds by up to 10x.
- Drove the integration of AMD's AI Engine (AIE1 and AIE2) models into the new infrastructure, exceeding the initial proof-of-concept scope by transpiling and fine-tuning a cycle-accurate model into a functional, untimed simulation model.
- Developed a comprehensive software debug framework tailored for AI workloads, providing robust tooling to troubleshoot and optimize Convolutional Neural Network (CNN) architectures such as ResNet and YOLO.
- Developed a custom memory model in Xilinx Vitis for advanced performance analysis of Alveo data center accelerator cards, supporting high-performance compute workloads across Machine Learning (ML) inference and FinTech algorithmic trading.
- Accelerated Xilinx Vitis hardware emulation speed by 6x (doubling the MVP target of 3x), significantly reducing simulation and debug turnaround times for complex architectures.
- Engineered a scalable, latency-accurate I/O emulation feature for the Xilinx Accelerated Algorithmic Trading (AAT) framework, enabling cycle-accurate simulations using PCAP inputs for precise packet-level visibility.

SAMSUNG

Bengaluru, India

Associate Staff Engineer

June 2016 – June 2019

- Architected a Cycle-Accurate Hybrid Simulation Framework that synchronized SystemC and FPGA environments, enabling seamless emulation of complex, distributed designs.
- Developed critical cycle-accurate models and transactors, including an AXI3 Interconnect SystemC model and AXI-to-TLM transactors, to drive advanced performance measurement for memory subsystems.
- Engineered a custom Co-Simulation framework leveraging Interconnect WorkBench (IWB/Incisive), SystemC, and NC-Sim to streamline cross-platform verification and design testing.

PATENTS

- **Model Level Debugging of Machine Learning Designs on Neural Processing Units** (Published, US-20260186951-A1)
- **Multi-Threaded Cycle Accurate Architecture Simulation** (Published, US20240184616-A1)
- **Data traffic injection for simulation of circuit designs** (Issued, US-11630935-B1)
- **Circuit design simulation and clock event reduction** (Published, US-20230114858-A1)
- **Adaptive Simulation Framework for Switching Between Functional and Cycle-accurate Execution in Real-Time** (Pending)

PUBLICATIONS

- Performance analysis of DS-CDMA using different chaotic sequences – IEEE (Mar 2016)

CERTIFICATIONS

- GPU Architectures and Programming – NPTEL (Topper) *May 2022*
- Multi-Core Computer Architecture – Storage and Interconnects – NPTEL (Elite Gold) *Oct 2022*
- Operating System Fundamentals – NPTEL (Topper) *Nov 2021*
- Modern C++ Concurrency in Depth – Udemy *Oct 2020*

SKILLS & AWARDS

- Languages & Libraries: C++, SystemC, Python
- Awards: AMD Executive Spotlight Award, Xilinx Quarterly Award, Samsung Employee of the Month
- Tools & Frameworks: Synopsys VDK, AMD Vivado / Vitis, AMD xsim, Cadence IWB / Incisive, NCSim

EDUCATION

M.Tech in Software Systems	BITS Pilani (WILP)
CGPA: 8.19	
B.Tech in Electronics & Communication Engineering	NIT Warangal
CGPA: 8.94	